

National University of Computer and Emerging Sciences
Lahore Campus

Probability & Statistics

MID-1 Exam

(MT2005)

Date: Feb 21, 2026

Course Instructor(s)

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Total Time (Hrs): 1

Total Marks: 35

Total Questions: 2

ALL SECTIONS BCS

Roll No

Section

Student Signature

Do not write below this line

- **Attempt all questions**
- Lead Pencil work wouldn't be marked or claimed for rechecking. Use permanent ink pen.
- If you found any ambiguity in the data then do not ask anything to the invigilator, just make an assumption and continue solving your paper.
- Solve Question no 2 on the Question paper For this specific question, solution on the answer sheet will not be considered.

CLO 1: Statistical Data Interpretation: Analyze and interpret various data types by computing measures of central tendency and dispersion, constructing frequency distributions, and utilizing graphical techniques for precise data representation.

Q1. Case Study: Program Performance Analysis

[Marks: 10+15]

A computer science laboratory conducted a controlled performance study on a sample program executed across Windows, macOS, and Linux operating systems. The purpose of the study was to examine how different operating environments influence program efficiency and resource utilization. The collected dataset records execution time (in milliseconds), CPU load (%), and memory usage (MB) for each run. Additionally, the program type (compilation or simulation) was included to capture variations in workload behavior.

Perform the following tasks for the provided data:

- Construct a cross-tabulation comparing execution time across operating systems to obtain meaningful insights from the data. Execution time should be grouped into classes starting at 170 ms, with a class width of 10 ms, to facilitate a clear and systematic comparison.

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b) The execution times (in milliseconds) for a sample program running on Windows and macOS are provided. Compare the two operating systems and determine which operating system demonstrates more reliable and consistent performance.

boxplot / CV

Run	Operating System	Execution Time (ms)	CPU Load (%)	Memory Usage (MB)	Program Type
1	Linux	179	52	890	Compilation
2	macOS	205	68	1120	Simulation
3	Windows	212	78	1220	Compilation
4	Linux	181	54	910	Simulation
5	macOS	195	60	1080	Compilation
6	Windows	219	85	1300	Simulation
7	Linux	178	50	880	Compilation
8	macOS	202	66	1110	Compilation
9	Windows	205	70	1150	Simulation
10	Linux	182	57	920	Simulation
11	macOS	198	63	1090	Simulation
12	Windows	210	75	1200	Compilation
13	Linux	180	55	900	Compilation
14	Windows	215	80	1250	Compilation
15	macOS	200	65	1100	Compilation

CLO 1: Statistical Data Interpretation: Analyze and interpret various data types by computing measures of central tendency and dispersion, constructing frequency distributions, and utilizing graphical techniques for precise data representation.

Q2:

[Marks:10]

A data analyst is working on a digital media analytics project to study the performance of top YouTube content creators. For this purpose, the analyst uses a manually curated dataset containing information on the top 100 YouTube channels in 2024.

The dataset includes the following variables:

- ❖ Number of subscribers
- ❖ Total views
- ❖ Total number of uploaded videos
- ❖ The official name of YouTube channels
- ❖ The country where the YouTube channel is based.
- ❖ Channel rank based on performance metric

The data represents a snapshot of channel performance as of December 2024 and covers creators from various genres, including entertainment, education, technology, and lifestyle. The dataset is compiled using publicly available YouTube statistics and verified third-party analytics platforms to ensure accuracy and transparency.

The analyst aims to explore:

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- ❖ Trends in audience growth
- ❖ Viewer engagement patterns
- ❖ Relationships between content volume and popularity

This analysis will help marketers, content creators, and data scientists understand platform dynamics and identify strategies used by the most influential YouTube creators

Scenario	Response
1 What type of study design does it demonstrate?	cross sectional study design. (1)
2 Identify the variable: "Rank of the channel based on performance metric", as qualitative or quantitative variable. Also indicate its measurement scale.	qualitative ordinal (2)
3 Which measure of central tendency would best represent for the following cases? A- Removing top 5% of channels avoids the few extremely popular channels inflating the average, B- The typical number of subscribers a channel has, ignoring extreme.	A: Trimmed mean (1) B: Interquartile range because it ignores the extreme
4 Suggest appropriate visualizations for analyzing: The country where the YouTube channel is based.	Tabular data where each YouTube channel has its corresponding country. → line plot.
5 Assuming that number of uploaded video follow the normal distribution. The mean number of uploaded videos is 400 with standard deviation is 50. If T-series has total number of uploaded video is 100. Identify whether T-series uploaded video consider to be an outlier?	$\frac{100 - 400}{50} \rightarrow -6$ (2) Since -6 is out of -3 to 3 it is outlier
6 Suppose the Five point summary of distribution of number of subscribers is calculated and the lower and upper limits to identify the anomalies are 5,000,000 subscribers and 45,000,000 subscribers respectively. The interquartile range is 10,000,000. Find the lower and upper quartiles.	$Q1 \rightarrow 20,000,000$ (1) $Q3 \rightarrow 60,000,000$

Lower =

$$5,000,000 = Q1 - 1.5 (10,000,000)$$

$$5,000,000 + 15,000,000 \rightarrow 20,000,000$$

higher

$$45,000,000 = Q3 + 1.5 (10,000,000)$$

$$Q3 = 60,000,000$$